

WIRELESS SERVO MOTOR CONTROL WITH ARDUINO

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ABSTRACT:

- The "Wireless Servo Motor Control with Arduino" project for remotely controlling servo motors using Arduino wireless communication technology.
- Servo motors are widely used in robotics, automation, and various other applications that require precise control over angular motion. This project offers a wireless solution to control servo motors, eliminating the need for physical connections and enabling remote operation.
- This project presents a wireless servo motor control system implemented with Arduino and the NRF24L01 wireless communication module.
- The NRF24L01 module is RF module.

PROPOSED METHOD :

Hardware Setup:

Assemble the hardware components, which include two Arduino boards (one for the sender and one for the receiver), NRF24L01 wireless communication modules, servo motors, and a power source (usually a battery or a power supply).

Wiring Connections:

Connect the NRF24L01 modules to each Arduino.

Connect the servo motor to the receiver Arduino for servo control.

Arduino Programming:

Write code for both the sender and receiver Arduinos.

In the sender Arduino code, include logic to read input from a user interface potentiometers this input to servo control commands.

Use the NRF24L01 library to set up wireless communication and transmit the control commands to the receiver

Receiver Code:

In the receiver Arduino code, configure the NRF24L01 module to receive data packets from the sender and use them to control the servo motor.

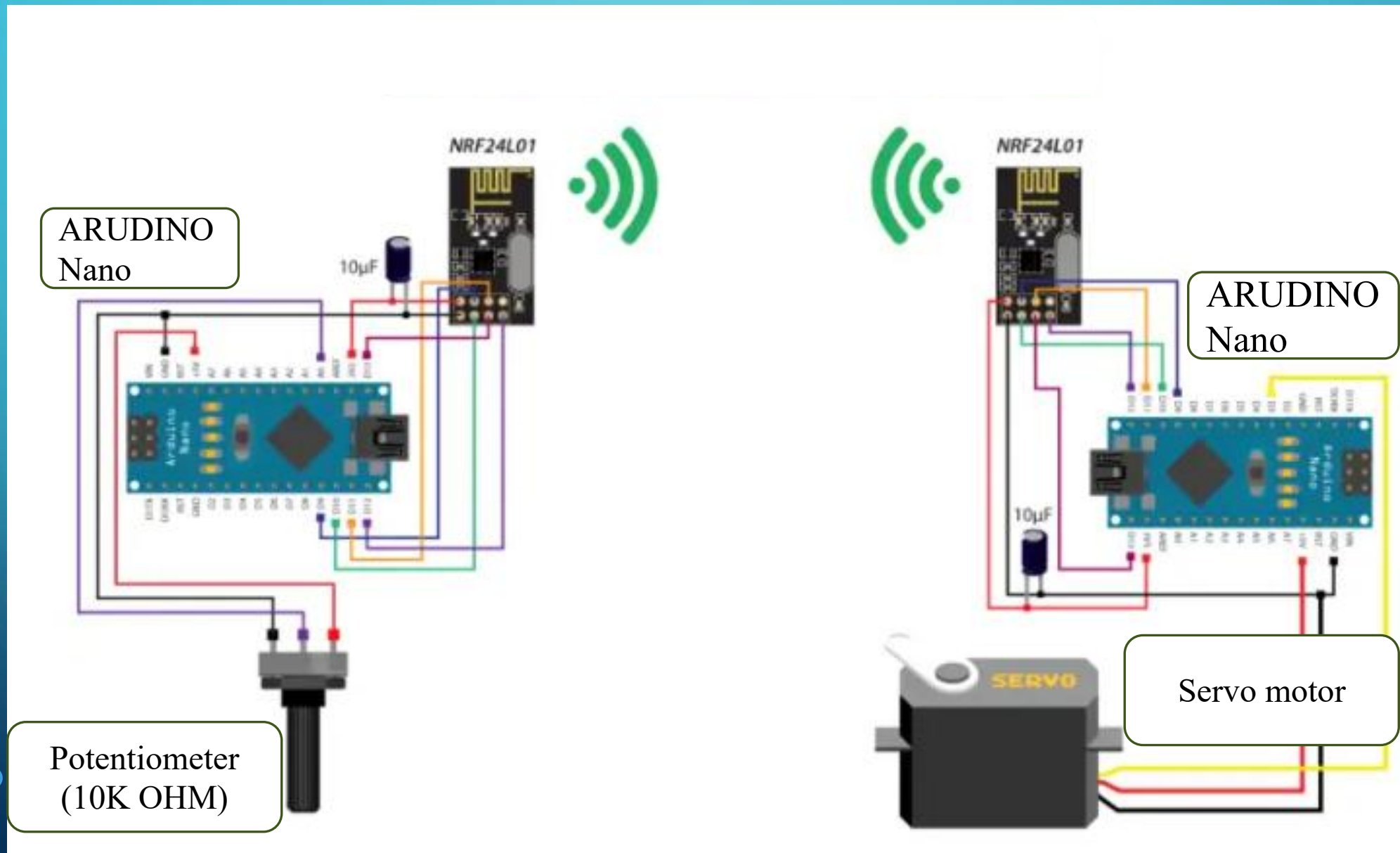
Servo Motor Control:

Ensure that the servo motor movement is smooth and accurate.

By following this we can create a wireless servo motor control system using Arduino and NRF24L01, making it suitable for various applications where remote and precise control of servo motors is required.

- ARUDINO

BLOCK DIAGRAM WITH PARAMETERS



A decorative graphic consisting of white circuit-like lines and circles on a blue gradient background, located on the left and right edges of the slide.

HARDWARE:

- NRF24L01 Module
- Potentiometer
- Arudino nano
- Servo motor
- Breadboard
- Jumper wire

SOFTWARE:

- Arduino Software (IDE)

The background is a blue gradient with decorative white circuit-like lines in the corners. The word "RESULT:" is centered in the upper left area.

RESULT:

REFERENCE

- <https://howtomechatronics.com/tutorials/arduino/how-to-build-an-arduino-wireless-network-with-multiple-nrf24l01-modules/>
- <https://youtu.be/xb7psLhKTMA?feature=shared>

The image features a vibrant blue background with a subtle, intricate pattern of circuit board traces and circular nodes, reminiscent of a microchip or a digital network. In the center, a solid black rectangle with rounded corners serves as a focal point. Inside this rectangle, the words "THANK YOU" are written in a classic, white, serif typeface, centered both horizontally and vertically. The overall aesthetic is clean, modern, and tech-oriented.

THANK YOU